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Critical Behaviors in a Stochastic One-Dimensional Sand-Pile Model*

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Abstract *A one-dimensional sand-pile model (Manna model), which has a stochastic redistribution process, is studied both in discrete and continuous manners. The system evolves into a critical state after a transient period. A detailed analysis of the probability distribution of the avalanche size and duration is numerically investigated. Interestingly, contrary to the deterministic one-dimensional sand-pile model, where multifractal analysis works well, the analysis based on simple finite-size scaling is suited to fitting the data on the distribution of the avalanche size and duration. The exponents characterizing these probability distributions are measured. Scaling relations of these scaling exponents and their universality class are discussed.*

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Key words: self-organized criticality, power-law, sand-pile model, finite-size scaling, universality class

1 Introduction

The concept of self-organized criticality (SOC) was proposed as a framework to understand the dynamics of extended dissipative systems by Bak, Tang, and Wiensenfeld.[1] Without the fine tuning of parameters, the systems can reach a steady state which is characterized by a power-law distribution of the size of avalanches. Bak *et al.* originally used a cellular automaton, now referred to as the BTW sand-pile model,[1] to illustrate their ideas.

Since then, a number of self-organized critical models, which define a number of "universal classes" depending on their values of exponents,[2−6] have been introduced. On the other hand, as for the one-dimensional (1D) systems, Kadanoff *et al.* have introduced a class of sand-pile models, where the avalanche process is deterministic, and it seems that they do not show simple finite-size scaling but multifractal scaling behaviors.[7,8]

A group of researchers in Oslo conducted experiments on real rice piles[9] and showed that under some conditions a real rice pile displays SOC behaviors. Later being inspired by these experiments, Christensen *et al.* proposed a model, hereafter called Oslo rice-pile model,[10] which shows a simple finite-size scaling in 1D for the avalanche size distribution. A good understanding of the Oslo model was achieved by Paczuski and Boettcher. They find that it belongs to the same universality class as interface depinning, and a train model for earthquakes.[11]

Here we study a stochastic one-dimensional (S1D) sand-pile model. The avalanche propagates by redistributing the energy on all the unstable sites to their nearest neighbors randomly and independently until all the sites become stable, just like a two-dimensional Manna model.[12] We employ a parallel updating scheme during the avalanches. In this model, the randomness is internal and inherent in the dynamics. This model is different from most previous ones, where the avalanche process is deterministic. It evolves into a critical state after a short transient time, and we study the critical behaviors in this state. Contrary to the 1D models in Refs. [7] and [8], we find that distributions for avalanche size and duration have a clear power-law behavior that lasts for a few decades and that the simple finite-size scaling assumption of probability distributions for avalanche size and duration works well for this model. In this paper we characterize the 1D sand-pile universality class by applying moment analysis techniques.[13−17] In the following we will show that both the discrete model and continuous model show well-defined avalanche size and duration exponents. They can be coherently described within a finite-size scaling (FSS) framework. But the results show that the discrete model and the continuous model are in different universality classes. The difference of this model with the rice pile model is that in the latter the updates always involve a simultaneous redistribution of energy to the left and right nearest neighbor, whereas it is random in this model.

2 Model

The 1D stochastic model we study here is defined as follows. Consider a one-dimensional grid of size L with open boundaries, labeled by integers $i = 1$ to L. The

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model is conservative and the energy is dissipated only at boundary sites. Each site i is assigned a dynamic variable $E(i)$, which represents some physical quantity such as energy, stress, etc.[18] In a critical height model a state is called stable if for all sites $E(i) < E_c$, where E_c is a threshold value. The evolution between stable states is by the following rules.[18]

(i) ***Deposition rule*** The systems are driven at the boundary and $E(1)$ is increased by some amount δE. When an unstable state is reached rule (2) is applied.

(ii) ***Toppling rule*** If $E(i) \geq E_c$, toppling takes place and energy is randomly distributed in the following way:

$$E(i) \to E(i) - \sum_e \Delta E(i)\,,$$

$$E(i+e) \to E(i+e) + \sum_e \Delta E(i)\,,$$

where e are a set of vectors from the site i to its two neighbors. As a result of the toppling, the dynamic variable in one or more of the neighbors may exceed the threshold. The toppling rule is then applied until a stable state is reached.[18] A time step is defined as the toppling of all the sites having $\boldsymbol{E}(i) \geq \boldsymbol{E}_c$, after the completion of the previous time step. The rate of deposition is slow enough that any avalanche will have ended before a new δE is added. The system is initialized by setting $\boldsymbol{E}(i) = 0$ for all sites. That means the system is to be built up from scratch.

It was shown before that the parameters δE and E_c are irrelevant to the scaling behavior, and the critical exponents depend only on the vector ΔE, to be termed relaxation vector.[18,19] For a 1D grid with toppling to nearest neighbors, it is of the form $\boldsymbol{\Delta E} = (\boldsymbol{E}_L, \boldsymbol{E}_R)$, where E_L and E_R are the amounts transferred to the left and right nearest neighbors, respectively. In this model some nearest neighbors are randomly chosen for toppling. A possible realization of this model can be specified by three relaxation vectors (1, 1), (2, 0), and (0, 2), each one applied with a probability of 1/3. In the continuous version of this model, for which $E_c = 1$ and $0 < \delta E < 1$(we take $\delta E = 0.2$), the relaxation vector is given by $(\boldsymbol{b}, \boldsymbol{b})$, $(2\boldsymbol{b}, 0)$, and $(0, 2\boldsymbol{b})$, where $\boldsymbol{b} = \boldsymbol{E}(i)/2$ and $\boldsymbol{E}(i)$ is the amount of energy in the unstable site before the toppling has occurred. Apparently, the site i will become empty after toppling. In the 1D BTW model, $E_c = 2$, $\delta E = 1$, and $\Delta E = (1, 1)$. This model displays only trivial behavior.[1] We have analyzed this stochastic model for both discrete and continuous manners by determining their critical exponents. We prove they are in different universality classes.

3 Theoretical and Numerical Results

After a short transient time, the system reaches a critical steady state with avalanches of activity distributed according to power laws. The size of an avalanche can be defined in two ways: the number of topplings s, and avalanche time duration t. The FSS hypothesis, usually assumed in SOC systems, states that the probability distribution of each of them can be described as a power-law with a cutoff

$$P(x) = x^{-\tau} F\Big(\frac{x}{x_c}\Big)\,, \tag{1}$$

where $x = s, t$, and F is a scaling function decaying rapidly for large arguments. When the system size L goes to infinity the cutoff x_c diverges as $x_c \propto L^{\beta_x}$. If the finite size scaling (FSS) assumption of Eq. (1) is valid, the set of exponents $\{\tau_x, \beta_x\}$ completely defines the universality class of the model.

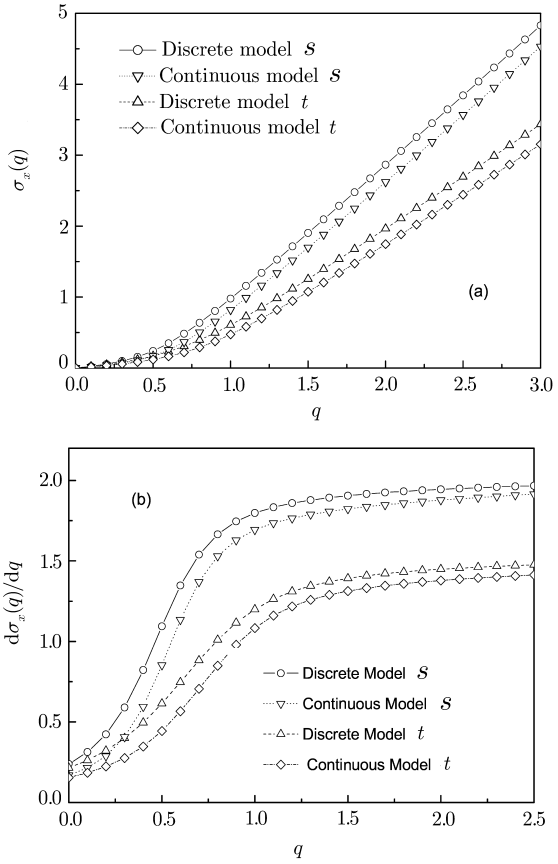


Fig. 1 (a) Plot of the momentum spectrum for the distribution of avalanche size s and time duration t. We get a clear linear behavior starting from $q \approx 1.3$. (b) Derivatives of the exponents $\sigma_x(q)$ for the rice-pile model that reach a saturated value at $q \approx 1.3$. This is not true for smaller q. For smaller q, we observe deviations from standard FSS, because the integral in Eq. (2) is dominated by its lower cutoff for small q moments.

The determination of the scaling exponents for the sand-pile models has often been performed in previous works by a direct measurement of the slope of a log-log plot of distributions.[20,21] But one has to mention that

the accuracy of the direct measurement is strongly affected by the upper and lower cutoffs of the distributions. We use more powerful moment analysis and data collapse techniques that use the whole set of data (not only the power law regime) and contain explicitly the system-size dependency. In the SOC field, the moment analysis has been introduced by De Menech *et al.* in the context of the two-dimensional BTW model.[13,14] Later it was successfully used on both Abelian and non-Abliean sand-pile models.[2,15−17]

Table 1 Comparisons of critical exponents for the S1D model with those for the rice-pile model. The values of β_s and β_t are obtained by measuring the slopes of the linear part in Fig. 1(a). The values of τ_s and τ_t are obtained according to Eq. (3) at $q = 2$.

Critical exponents	τ_s	β_s	τ_t	β_t
Discrete model	1.53 ± 0.01	1.95 ± 0.01	1.65 ± 0.01	1.45 ± 0.01
Continuous model	1.59 ± 0.01	1.86 ± 0.01	1.72 ± 0.01	1.36 ± 0.01
Rice-pile model	1.53 ± 0.05[19]	2.20 ± 0.05[19]	1.84 ± 0.05[19]	1.40 ± 0.05[19]

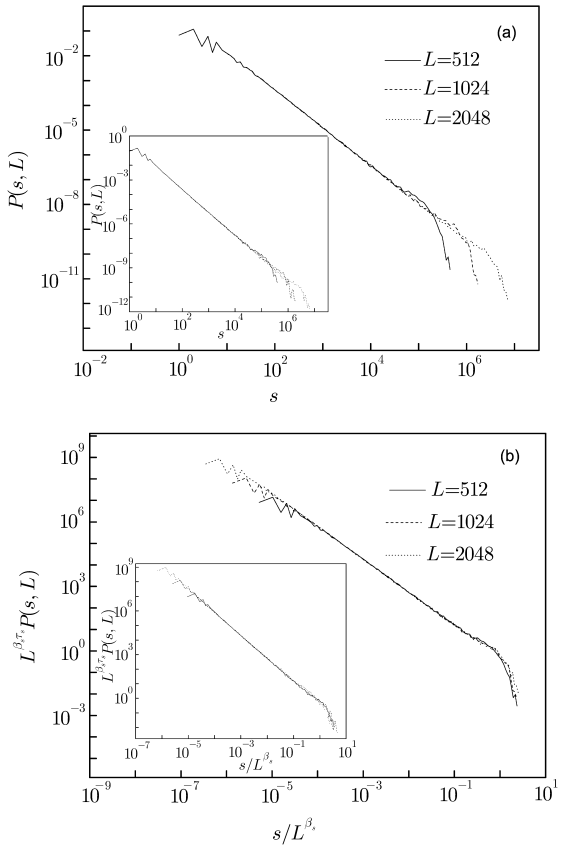


Fig. 2 (a) Log-log plot of the avalanche size probability distribution for several values of the system size L in the discrete case. The data have been binned over exponentially increasing sizes with base 1.1. In the inset it is shown for the continuous case. (b) Data collapse analysis of the curves shown in Fig. 2(a) for the avalanche size distribution (the values used for the critical exponents are $\tau_s = 1.53$ $\beta_s = 1.95$ in the discrete case, and $\tau_s = 1.59$ $\beta_s = 1.86$ in the continuous case).

In order to test the above FSS picture and to find an accurate estimate of the various critical exponents, we perform numerical simulation of this model with the system sizes from $L = 512$ to $L = 2048$, and the statistical distributions are obtained averaging over 10^6 nonzero avalanches for each system size. We define the $q-$moment of x on a lattice of size L as $\langle x^q \rangle_L = \int_1^\infty x^q P(x) \mathrm{d}x$. If FSS hypothesis (Eq. (1)) is valid, at least in the asymptotic limit ($x \to \infty$), we can transform $z = x/L^\beta$ and obtain

$$\langle x^q \rangle_L = L^{\beta_x(q+1-\tau_x)} \int x^{q+\tau_x} F(z) \mathrm{d}z \sim L^{\beta_x(q+1-\tau_x)} . \quad (2)$$

More generally, we can write $\langle x^q \rangle_L \sim L^{\sigma_x(q)}$, where the exponents $\sigma_x(q)$ can be obtained as the slope of the log-log plot of $\langle x^q \rangle_L$ as a function of L. Comparing with Eq. (2), we have $\langle x^{q+1} \rangle_L / \langle x^q \rangle_L \sim L^{\beta_x}$ or $\sigma_x(q+1) - \sigma_x(q) = \beta_x$, so that the slope of $\sigma_x(q)$ versus q is the cutoff exponent; i.e., $\beta_x = \partial \sigma_x(q)/\partial q$. For very small q, this is not true, because the integral in Eq. (2) is dominated by its lower cutoff for small q moments. Once the exponent β_x is known, we can calculate τ_x from the scaling relationship

$$(q + 1 - \tau_x)\beta_x = \sigma_x(q) , \quad (3)$$

which should be satisfied for large enough sizes.

We have calculated the moments $\sigma_x(q)$ for our data from both discrete and continuous cases. In the presence of FSS assumption, as we have argued above, one should expect to observe a linear behavior for $\sigma_x(q)$, namely $\mathrm{d}\sigma_x(q)/\mathrm{d}q = \beta_x \equiv$ const. Simulations on the Manna model, as in Ref. [17], show that it is indeed the case, with the moments $\sigma_x(q)$ that display a clear linear behavior and a slope for the moments $\sigma_x(q)$ that reaches a saturation value for relatively small values of q. Our moment analysis yields good results for the avalanche size and duration distributions, with the moments $\sigma_x(q)$ that display a clear linear behavior starting from small q and derivatives of the moments $\sigma_x(q)$ that reach a saturation

plateau for small values of q (see Fig. 1). Figures 1(a) and 1(b) show the avalanche size and duration distribution exponents for the two types of models. We can clearly see that the scaling exponents of the continuous case are quite different from those of the discrete case. We can obtain the values of β_s and β_t by measuring the slopes of the linear part in Fig. 1(a). The values of τ_s and τ_t are obtained according to Eq. (3) at $q = 2$. As shown in Table 1, this model and the rice pile model have different exponents, thus we conclude that this model and the Oslo rice pile model are not in the same universality class.

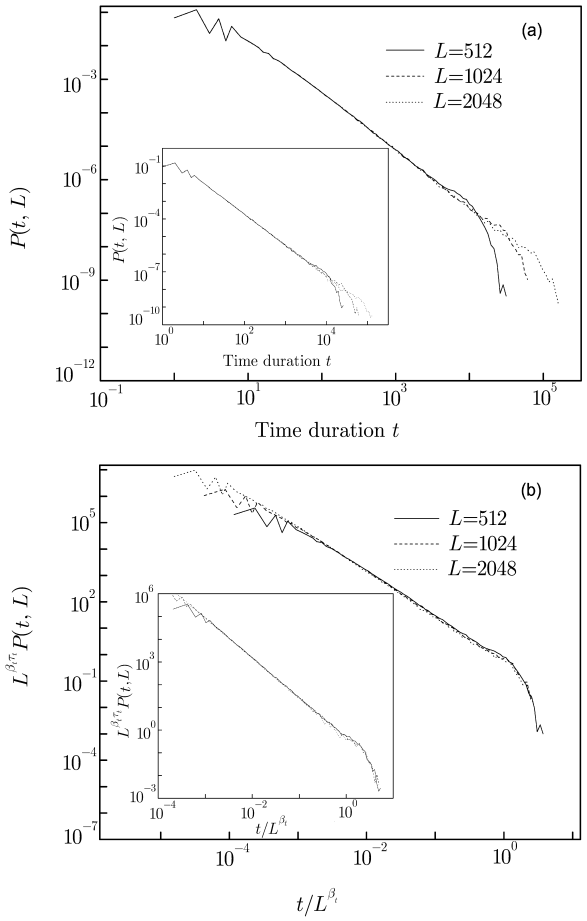


Fig. 3 (a) Log-log plot of the avalanche time duration probability distribution for several values of the system size L in the discrete case. The data have been binned over exponentially increasing sizes with base 1.1. In the inset it is shown for the continuous case. (b) Data collapse analysis of the curves shown in Fig. 3(a) for the avalanche time duration distribution (the values used for the critical exponents are $\tau_t = 1.65$ and $\beta_t = 1.45$ in the discrete case, and $\tau_t = 1.72$ and $\beta_t = 1.36$ in the continuous case).

We can check the validity of this result by means of a data collapse technique: if the FSS picture (Eq. (1)) is correct, by rescaling $q \equiv x/L^{\beta_x}$ and $P_q \equiv P(x, L)L^{\beta_x\tau_x}$, the data for different L must then collapse onto the same universal curve. In Fig. 2(a), the distribution is plotted in a log-log scale for different system sizes L in the case of continuous model. Apparently the distribution is a power law with the presence of a peak close to the cutoff size $s_c \propto L^{\beta_s}$. In the inset of Fig. 2(a), the distribution is plotted in a double logarithmic scale for different system sizes L in the case of continuous model. In Fig. 2(b) we plot the data collapse analysis for the avalanche size distribution. In Fig. 3(a), the distribution of avalanche time duration t is plotted in a log-log scale for different system sizes L in the case of discrete model. Again we observe that a power-law distribution with the presence of a peak close to the cutoff size $t_c \propto L^{\beta_t}$. In the inset of Fig. 3(a), the same is obtained for different system sizes L in the case of continuous model. We also see that increasing L, the region of the power law increases. But we can observe that the region of power law of the avalanche durations distribution is about two or three decades less than that of the avalanche size distribution. It is due to that the finite size effect is more obvious on the avalanche durations distribution than on the avalanche size distribution. In Fig. 3(b), we get a good data collapse with the exponents we have obtained above. The perfect collapse of Figs. 2(b) and 3(b) confirms the good scaling as we argued above.

According to Eq. (1), from conservation of the probability follows the exponents scaling relation[11]

$$\beta_t(\tau_t - 1) \equiv \beta_s(\tau_s - 1)\,. \tag{4}$$

Our results agree well with this scaling relation.

4 Conclusion

In summary, we have studied an S1D sand-pile model. In this system, the randomness in the relaxation process makes the model different from the D1D sand-pile model, where the randomness is only external. In this model, the randomness is internal and inherent in the dynamics. We have given a detailed numerical analysis of critical behavior in this model. Applying the moment analysis technique, we have studied scaling behaviors of the probability distributions for avalanche size and duration. We find that the analysis based on simple finite size scaling is suitable to fitting the data on the distribution of the avalanche size and duration. In the discrete model and continuous model with stochastic relaxation process, the moment analysis of the avalanche size (s or t) distribution yields a simple linear dependence of the exponents $\sigma_x(q)$ and this model shows well-defined size and duration exponents. The origin of the good scaling behavior can be attributed to the internal and inherent randomness in the dynamics. In the discrete manner, the self-organized critical model belongs to the universality class that is characterized by the avalanche exponents $\beta_s = 1.95$ and $\beta_t = 1.45$, whereas the model with continuous mechanism belongs to a new universality class characterized by $\beta_s = 1.86$ and $\beta_t = 1.36$.

And also it is numerically shown that the discrete model and the rice-pile model have different exponents, thus we conclude that the discrete model and the rice-pile model are not in the same universality class. Work is in the progress to obtain comparable results for two-dimensional and three-dimensional cases.

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